

Chapter 3

Continuous System Simulation

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3.1 Continuous system

- A continuous system is one in which the state variable(s) change continuously over time. E.g. the amount of water flow over a dam.
- A continuous system is the one in which predominant activities of the system cause smooth changes in the attributes of the system entities.
- When such a system is modeled mathematically, the variables of the model representing the attributes are controlled by continuous functions.
- Generally, in continuous systems, the relationships describe the rates at which attributes change, so that the model consists of differential equations.
- The simplest differential equation model with one or more linear equations with constant coefficients can be solved analytically without the use of simulation.
- However, when nonlinearities are introduced into the model, it frequently becomes impossible or, at least, very difficult to solve the models.
- The method of applying simulation to continuous models can therefore be developed by showing their application to models where the differential equations are linear and have constant coefficients, and then generalizing to more complex equations.

3.1 Continuous system

- An example of differential equation with constant coefficients can be taken as:

$$M\ddot{x} + D\dot{x} + Kx = KF(t)$$

- Here, the dependent variable x appears together with its first and second derivatives and that the terms involving these quantities are multiplied by constant coefficients and added.
- The quantity $F(t)$ is an input to the system, depending upon the independent variable t .
- If the dependent variable or any of its derivatives appear in any other form, such as being raised to a power, or are combined in any other way for example, by being multiplied together, the differential equation is said to be nonlinear.
- When more than one independent variable occurs in a differential equation, the equation is said to be partial differential equation.
- It can involve the derivatives of the same dependent variable with respect to each of the independent variables.

3.1 Continuous system

- Differential equations both linear and non-linear, occur repeatedly in scientific and engineering studies. The reason for this prominence is that most physical and chemical processes involves rates of change, which require differential equations for their mathematical description.
- Therefore, **Continuous System Simulation** refers to a computer model of a physical system that continuously tracks system response according to a set of equations typically involving differential equations
- Continuous System simulation allows prediction of:
 - rocket trajectories
 - hydrogen bomb dynamics
 - electric circuit simulation
 - human respiration
 - robotics

3.1 Example: Continuous System

Distributed Lag Models

- Models that have the properties of changing only at fixed intervals of time, and of basing current values of the variables on other current values and values that occurred in previous intervals, are called distributed lag models.
- They are used extensively in econometric studies where the uniform steps correspond to a time interval, such as a month or a year, over which some economic data are collected.
- These models consist of linear algebraic equations.
- They represent a continuous system, but one in which the data is only available at fixed points in time.

Distributed Lag Models-Contd..

- As an example, consider the following simple mathematical model of the national economy:
- Let, C be consumption, I be investment, T be taxes, G be government expenditure and Y be national income.

$$C=20+0.7(Y-T)$$

$$I=2+0.1Y$$

$$T=0.2Y$$

$$Y=C+I+G$$

- All quantities are expressed in billions of dollars.
- This is a static mathematical model, but it can be made dynamic by picking a fixed time interval, say one year, and expressing the current values of the variables in terms of values at previous intervals.

Distributed Lag Models-Contd..

- Any variable that appears in the form of its current value and one or more previous intervals is said to be a lagged variable.
- Its value in a previous interval is denoted by attaching the suffix $-n$ to the variable, where n indicates the interval, with 1 denoting the previous interval, 2 denoting the one prior to that, and so on.
- In this example we will only go back one interval such as:

$$I = 2 + 0.1Y_{-1}$$

$$T = 0.2Y_{-1}$$

$$Y = C_{-1} + I_{-1} + G_{-1}$$

$$C = 20 + 0.7(Y_{-1} - T_{-1})$$

Distributed Lag Models-Contd..

- Given an initial set of values for all variables, the values of the variables at the end of one year can be derived.
- Taking these values as the new values of the lagged variables, the values can be derived at the end of the second year and so on.
- There are only four equations in five unknowns, so one variable must be specified for each interval.

3.2 Analog Computer

- An analog computer is a computation device that uses continuously variable physical phenomena such as electrical voltages, mechanical motion, or hydraulic pressure to model and solve mathematical problems.
- The most widely used form of analog computer is the electronic analog computer, based on the use of high gain dc (direct current) amplifiers, called operational amplifiers.
- Voltages in the computer are equated to mathematical variables, and the operational amplifiers can add and integrate the voltages.
- With appropriate circuits, an amplifier can be made to add several input voltages, each representing a variable of the model, to produce a voltage representing the sum of the input variables.

3.2 Analog Computer

- Different scale factors can be used on the inputs to represent coefficients of the model equations.
- Another circuit arrangement produces an integrator for which the output is the integral with respect to time of a single input voltage or the sum of several input voltages.
- All voltages can be positive or negative to correspond to the sign of the variable represented.
- To satisfy the equations of the model, it is sometimes necessary to use a sign inverter, which is an amplifier designed to cause the output to reverse the sign of the input.

3.2 Analog Computer

- Electronic analog computers are limited in accuracy for several reasons.
 - It is difficult to carry the accuracy of measuring a voltage beyond a certain point.
 - Secondly, a number of assumptions are made in deriving the relationships for operational amplifiers, none of which is strictly true; so, amplifiers do not solve the mathematical model with complete accuracy.
 - A particularly troublesome assumption is that there should be zero output for zero input.
 - Another type of difficulty is presented by the fact that the operational amplifiers have a limited dynamic range of output, so that scale factors must be introduced to keep within the range.
- As a consequence, it is difficult to maintain an accuracy better than 0.1% in an electronic analog computer.

3.2 Analog Computer

- A digital computer is not subject to the same type of inaccuracies.
- Virtually any degree of accuracy can be programmed and, with the use of floating-point representation of numbers, an extremely wide range of variations can be tolerated.
- Integration of variables is not a natural capability of a digital computer, as it is in an analog computer, so that integration must be carried out by numerical approximations.
- However, methods have been developed which can maintain a very high degree of accuracy.
- A digital computer also has the advantage of being easily used for many different problems.
- An analog computer must usually be dedicated to one application at a time, although time-sharing sections of an analog computer has become possible.

3.2 Analog Computer

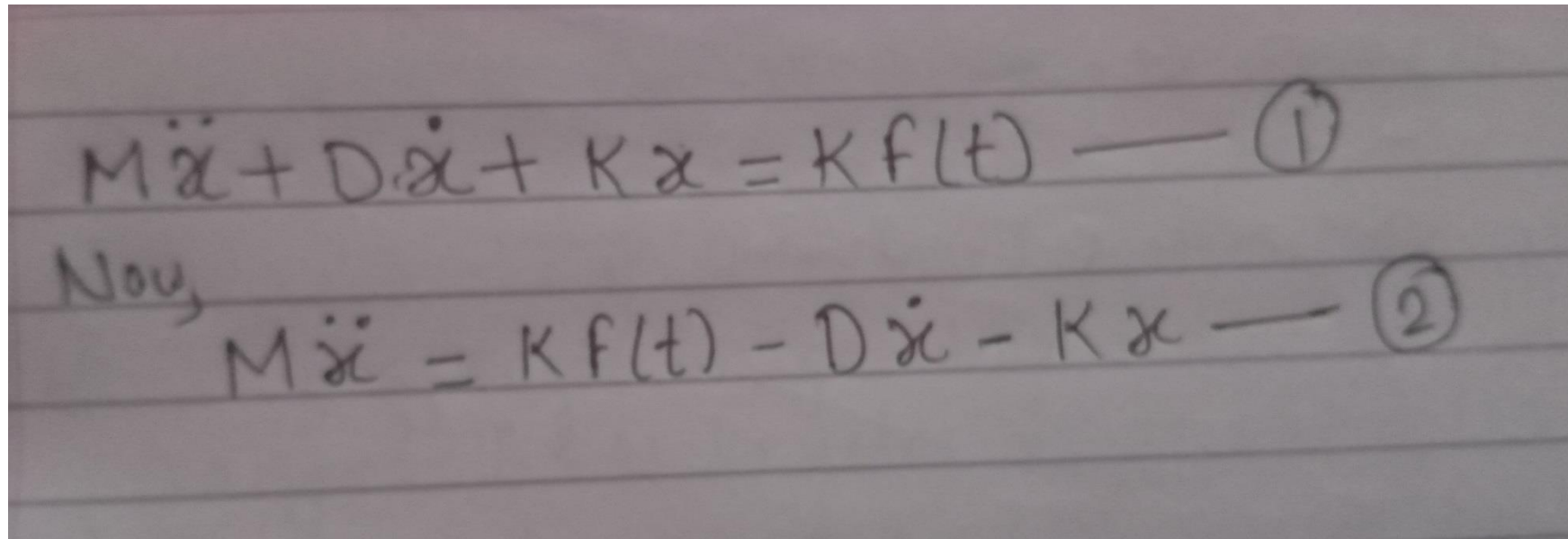
- In spite of the widespread availability of digital computers, many users prefer to use analog computers. There are several considerations involved.
 - The analog representation of a system is often more natural in the sense that it directly reflects the structure of the system; thus simplifying both the setting up of a simulation and the interpretation of the results.
 - Under certain circumstances, an analog computer is faster than a digital computer, principally because it can be solving many equations in a truly simultaneous manner; whereas a digital computer can be working only on one equation at a time, giving the appearance of simultaneity by interfacing the equations.
 - On the other hand, the possible disadvantages of analog computers, such as limited accuracy and the need to dedicate the computer to one problem, may not be significant.

Assignment

- Differentiate between analog and digital computer.

3.3 Analog methods

- The general method by which analog computers are applied can be demonstrated using the second order differential equation that has already been discussed:



The image shows two handwritten equations on lined paper. The first equation is $M\ddot{x} + D\dot{x} + Kx = Kf(t)$ labeled with a circled 1. The second equation, preceded by the word 'Now,' is $M\ddot{x} = Kf(t) - D\dot{x} - Kx$ labeled with a circled 2.

$$M\ddot{x} + D\dot{x} + Kx = Kf(t) \text{ — (1)}$$

Now,

$$M\ddot{x} = Kf(t) - D\dot{x} - Kx \text{ — (2)}$$

3.3 Analog methods

- Suppose a variable representing the input $F(t)$ is supplied, and assume for the time being that there exist variables representing $-x$ and $-\frac{dx}{dy}$.
- These three variables can be scaled and added with a summer to produce a voltage representing $M\partial^2 x/\partial y^2$.
- Integrating this variable with a scale factor of $1/M$ produces $\frac{dx}{dy}$.
- Changing the sign produces $-\frac{dx}{dy}$ which supplies one of the variables initially assumed; and a further integration produces $-x$, which was the other assumed variable.
- For convenience, a further sign inverter is included to produce $+x$ as an output.
- A block diagram to solve the problem in this manner is shown in following Figure.

3.3 Analog methods

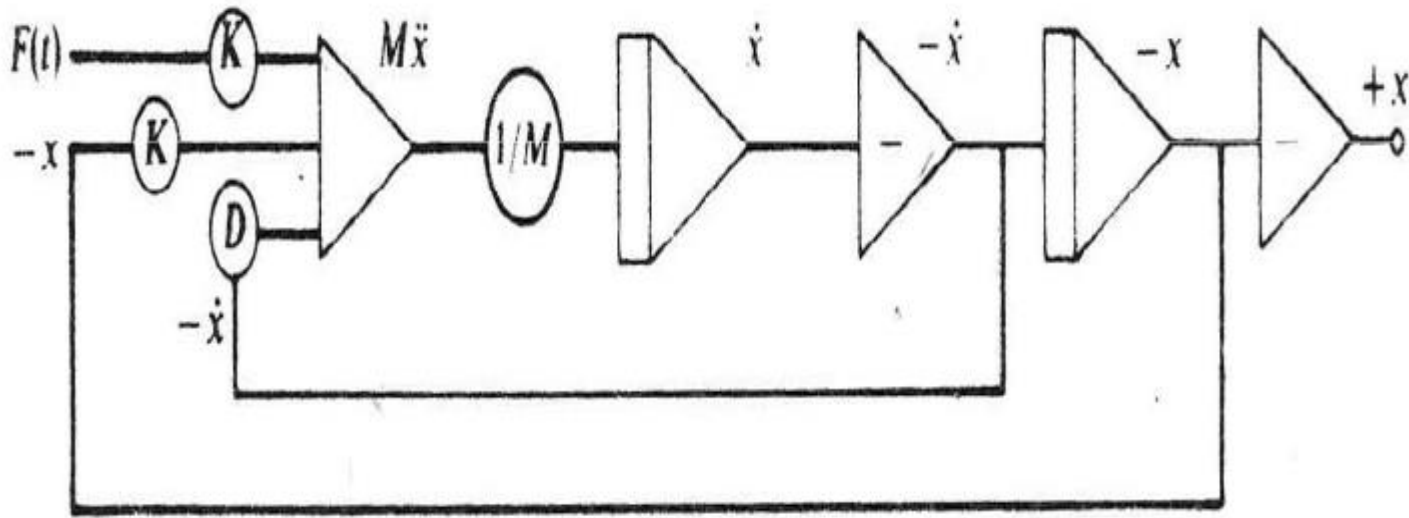


Diagram for the automobile suspension problem

3.3 Analog methods

- The symbols used in the figure are standard symbols for drawing block diagrams representing analog computer arrangements.
- The circles indicate scale factors applied to the variables.
- The triangular symbol at the left of the figure represents the operation of adding variables.
- The triangular symbol with a vertical bar represents an integration, and the one containing a minus sign is a sign changer.

3.3 Analog methods

- The addition on the left, with its associated scaling factors, corresponds to the addition of the variables representing the three forces on the wheel, producing a variable representing M .
- The scale is changed to produce $\partial^2 x / \partial y$ and the result is integrated twice to produce both $\dot{x}$ and x .
- Sign changers are introduced so that variables of the correct sign can be fed back to the adder, and the output can be given in convenient form.
- With an electronic analog computer, the variables that have been described would be voltages, and the symbols would represent operational amplifiers arranged as adders, integrators, and sign changers.

3.3 Analog methods

- The above Figure would then represent how the amplifiers are interconnected to solve the equation.
- It should be pointed out, however, that there can be several ways of drawing a diagram for a particular problem, depending upon which variables are of interest, and on the size of the scale factors.
- When a model has more than one independent variable, a separate block diagram is drawn for each independent variable and, where necessary, interconnections are made between the diagrams.
- As an example, the following Figure shows a block diagram for solving the model of the liver.

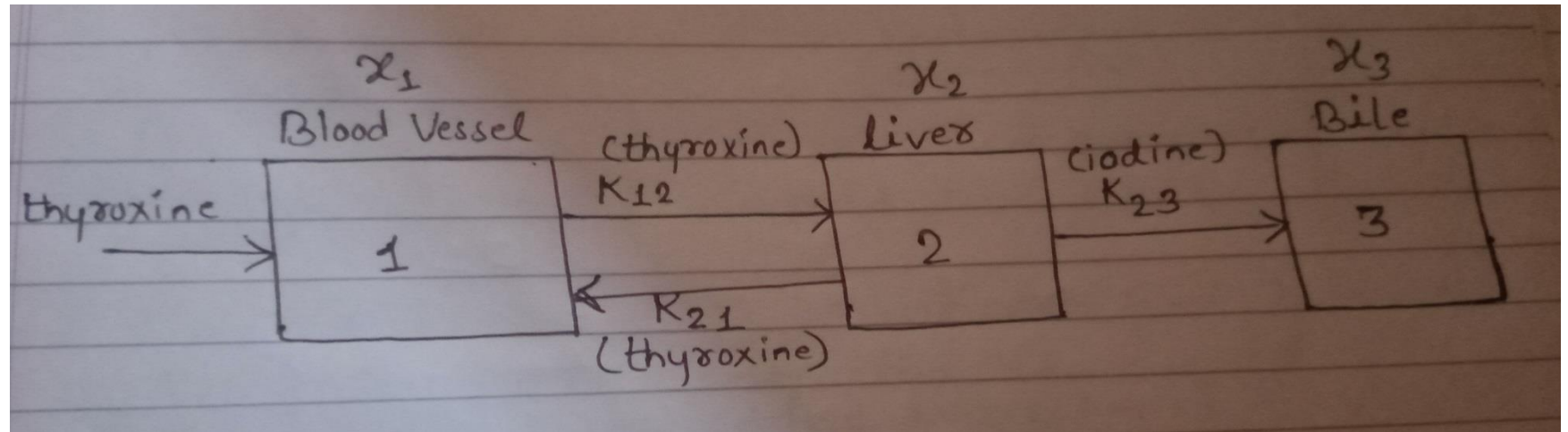
3.3 Analog Methods

- **Model of a Liver**

- The function of liver in human body can be explained as; when a chemical thyroxine is injected into the blood stream, it is carried to liver.
- The liver can change the thyroxine into iodine which is absorbed into the bile.
- However, neither the conversion nor the absorption occur instantaneously. Some of the thyroxine reenters the blood stream to be re-circulated and returned to the liver.
- To study this phenomenon, a mathematical model was constructed assuming that the body can be represented as 3 compartments and that the rates at which thyroxine is transferred between the compartments are proportional to the concentration of the thyroxine in the compartments.

3.3 Analog Methods

- The three compartments 1, 2 and 3 represents the blood vessels , the liver and bile respectively. The model leads to three differential equations as:

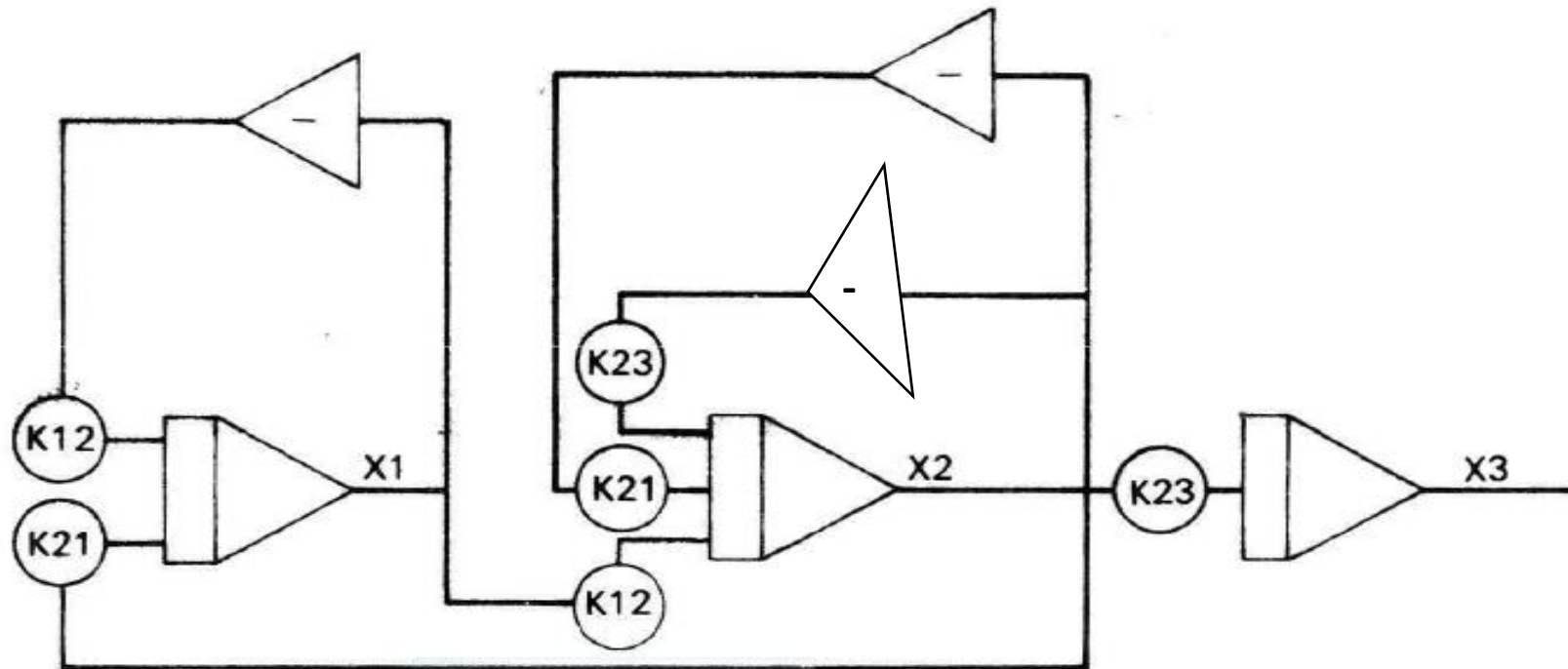


$$\frac{dx_1}{dt} = -k_{12}x_1 + k_{21}x_2$$

$$\frac{dx_2}{dt} = k_{12}x_1 - (k_{21} + k_{23})x_2$$

$$\frac{dx_3}{dt} = k_{23}x_2$$

3.3 Analog methods



Analog computer model of the liver

3.3 Analog methods

- There are three integrators, shown at the bottom of the figure. Reading from left to right, they solve the equations for X1, X2, X3.
- Interconnections between the three integrators, with sign changers where necessary, provide inputs that define the differential coefficients of the three variables.
- The first integrator, for example, is solving the equation:

$$\frac{dx_1}{dy} = -k_{12}x_1 + k_{21}x_2$$

3.3 Analog methods

- The second integrator is solving the equation:

$$\frac{dx_2}{dt} = k_{12}x_1 - (k_{21} + k_{23})x_2$$

- The last integrator solves the equation

$$\frac{dx_3}{dy} = k_{23}x_2$$

3.4 Hybrid Simulation

- Hybrid simulation is achieved by combining analog and digital computers.
- An approach to simulation in which a continuous simulation clock is maintained and updates to the model are event-driven.
- The form taken by hybrid simulation depends upon the application.
- One computer may be simulating the system being studied, while the other is providing a simulation of the environment in which the system is to operate.
- It is also possible that the system being simulated is an interconnection of continuous and discrete subsystems.
- High-speed converters are needed to transform signals from one form of representations to the other.

3.5 Digital-Analog Simulators

- Digital Analog Simulation (DAS) is a programming technique which makes a digital computer operate much like an analog computer.
- Digital-Analog Simulators are the simulators that allow a continuous model to be programmed on a digital computer.
- Languages contain macro-instructions that carry out the action of adders, integrators and sign changers.
- A program, is written to link together these macro-instructions.

3.6 Continuous System Simulation Languages (CSSLs)

- CSSLs use the statement-type of input for digital computers, allowing a problem to be programmed directly from the equations of mathematical model.
- CSSL uses macros and subroutines, that perform the function of specific analog elements.
- CSSL also allows the user to define special purpose elements that correspond to operations that are particularly important in specific types of applications.
- CSSL include a variety of algebraic and logical expressions.
- One particular CSSL described here is Continuous System Modeling Program, Version III, (CSMP III)

CSMP III

- A CSMP III program is constructed from three general types of statements:
 1. Structural Statements
 2. Data statements
 3. Control statement

CSMP III - Statements

Structural Statements:

- Structural statements which define the model. They consist of FORTRAN-like programming language statements, and functional blocks of program code (procedures) designed for repeat operations that frequently occur in a model definition.
- Structural statements can make use of the operations of addition, subtraction, multiplication, division and exponentiation, using the same notation and rules as used in FORTRAN.
- For e.g., the model includes the equation:

$$X = \frac{6Y}{w} + (Z-2)^2$$

The following statement would be used as:

$$X = 6.0*Y/W + (Z-2.0)**2.0$$

CSMP III - Statements

- There are many functional blocks which, in addition to providing operations specific simulation, duplicate many of the mathematical-function subprograms of FORTRAN.
- Among these are the exponential functions, trigonometric functions and functions for taking a maximum and minimum values.
- Below is the list of eleven of the functional blocks:

CSMP III - Statements

Functional Block	General Form
INTEGRATOR	$Y = \text{INTGRL} (IC, X)$ $Y(0) = IC$
LIMITER	$Y = \text{LIMIT} (P_1, P_2, X)$
STEP FUNCTION	$Y = \text{STEP} (P)$
EXPONENTIAL	$Y = \text{EXP} (X)$
NATURAL LOGARITHM	$Y = \text{ALOG} (X)$
TRIGONOMETRIC SINE	$Y = \text{SIN} (X)$
TRIGONOMETRIC COSINE	$Y = \text{COS} (X)$
SQUARE ROOT	$Y = \text{SQRT} (X)$
ABSOLUTE VALUE	$Y = \text{ABS} (X)$
LARGEST VALUE	$Y = \text{AMAX1} (X_1, X_2, \dots, X_n)$
SMALLEST VALUE	$Y = \text{AMIN1} (X_1, X_2, \dots, X_n)$

CSMP III - Statements

Data Statements:

- Data statements are those which assign numerical values to various changing parameters, constants, and initial conditions.
- Of the data statements, one called INCON can be used to set the initial values of the integration-function block.

Control Statements:

- Control Statements are those which specify options in assembly and execution of the program, and the choice of output of the results of the calculations performed.
- **CONST** = parameter can be given a value for a specific run with CONST control statement.
E.g. CONST A = 0.5, XDOT = 1.25
- **PARAM** = it can be used to assign values to individual parameters, but its chief purpose is to specify a series of values for one parameter only.
E.g. PARAM D = (0.25, 0.50, 0.75, 1.0)

CSMP III - Statements

Control Statements Contd...

- **PRINT** = gives the output in the form of printed tables.
- **PRTPLT** = gives the output in the form of print plotted graphs.
- **TIMER** = used to specify certain time intervals. An integration time size must be specified, a total simulation time must also be given and interval sizes for printing and plotting results needs to be specified. The following is an example:

TIMER DELT = 0.005, FINTIM = 1.5, PRDEL = 0.1, OUTDEL = 0.1

The items specified are:

DELT = Integration time

FINTIM = Finish time

PRDEL = Interval at which to print the result

OUTDEL = Interval at which to print-plot

CSMP III - Statements

Control Statements Contd...

- TITLE = can be used to put headings on the printed output
- LABEL = can be used to put headings on the print-plotted outputs
- END = it specifies the end of the all the three types of statements.

CSMP III -Example

Example for automobile suspension system

```
TITLE AUTOMOBILE SUSPENSION SYSTEM
  PARAM D = (5.656, 16.968, 39.592, 56.56, 113.12)
  X2DOT = (1.0/M) * (K*F - K*X - D*XDOT)
  XDOT = INTGRL(0.0, X2DOT)
  X = INTGRL(0.0, XDOT)
  CONST M = 2.0, F = 1.0, K = 400.0

  TIMER DELT=0.005, FINTIM=1.5, PRDEL=0.05, OUTDEL=0.05
  PRINT X, XDOT, X2DOT
  PRTPLT X
  LABEL DISPLACEMENT VERSUS TIME
END
STOP
```

3.7 Feedback System

- The system that takes feedback from the output i.e. input is coupled with output.
- Example a home heating system controlled by a thermostat is a simple example of a feedback system.
- The system has a furnace whose purpose is to heat a room and the output of the system can be measured as a room temperature.
- Depending upon whether the temperature is below or above the thermostat setting, the furnace will be turned on or off, so that the information is being feedback from the output to the input.
- Here, there are only two states, either the furnace is “on” or “off”.

References

G. Gorden, “ System simulation”